

Problem

Determine if these functions are even, odd, or neither.

(a) $1 + t$

(b) $t^2 - 1$

(c) $\cos n\pi t \sin n\pi t$

(d) $\sin 2\pi t$

(e) e^{-t}

Step-by-step solution

Step 1 of 5

(A) $1 - t \quad f(t) = (1 - t)$

$$f(-t) = 1 + t$$

$$f(t) \neq -f(t)$$

$$\text{Neither } R_1 \quad f(t) = f(t)$$

$f(t)$ is neither R_1 Odd nor Even.

Step 2 of 5

(B) $t^2 - 1 \quad f(t) = t^2 - 1$

$$f(-t) = t^2 - 1$$

$$f(-t) = f(t)$$

$f(t)$ is Even function.

Step 3 of 5

(C) $f(t) = \cos n\pi t \sin n\pi t.$

$$h(t) = \cos n\pi t \sin n\pi t$$

$$h(-t) = \cos(-n\pi t) \cdot \sin(-n\pi t)$$

$$h(-t) = \cos n\pi t \sin n\pi t$$

$$h(t) = -h(t)$$

$h(t)$ is odd function.

Step 4 of 5

(D) $f(t) = \sin^2 \pi t$

$$g(-t) = \sin^2(-\pi t)$$

$$= \sin^2(\pi t)$$

$$g(-t) = g(t)$$

$g(t)$ is Even function.

Step 5 of 5

(E) $y(t) = e^{-t}$

$$y(-t) = e^t$$

$y(t)$ is neither Even nor Odd.